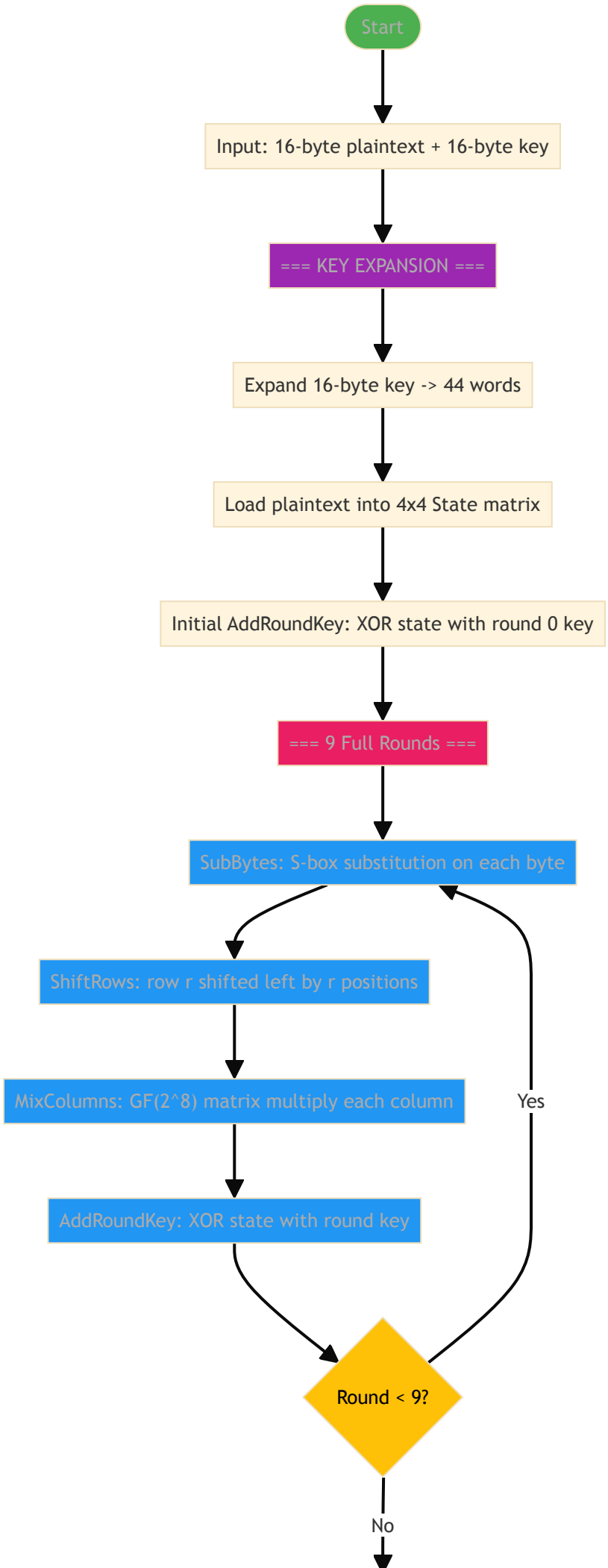
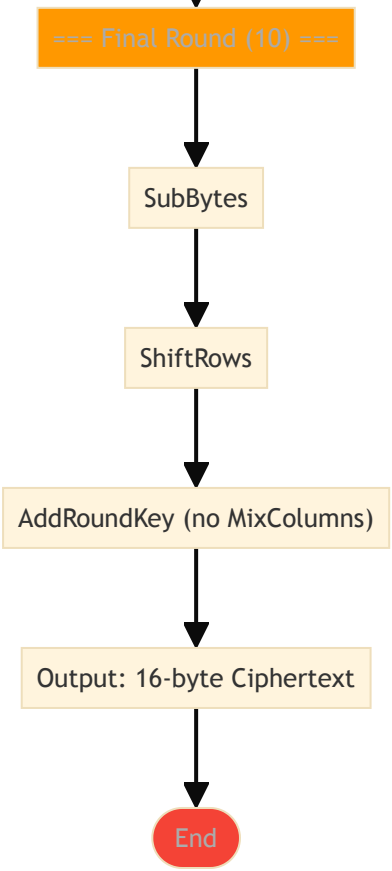

Start
Input: 16-byte plaintext + 16-byte key
=== KEY EXPANSION ===
Expand 16-byte key -> 44 words
Load plaintext into 4x4 State matrix
Initial AddRoundKey: XOR state with round 0 key
=== 9 Full Rounds ===
SubBytes: S-box substitution on each byte
ShiftRows: row r shifted left by r positions
MixColumns: GF(2^8) matrix multiply each column
AddRoundKey: XOR state with round key
Round < 9?
Yes
No


=== Final Round (10) ===
SubBytes
ShiftRows
AddRoundKey (no MixColumns)
Output: 16-byte Ciphertext
End